LEF MODEL · CHAPTER 19

Chapter 19: Young Mars — the Pointed Bulge of Tharsis and the Ejection of Deimos and Phobos from Pavonis Mons

Based on Paper 10 — Mars: The Protrusion and the Ring (Parts A–E: the expanded, fast Mars, the pointed bulge, Pavonis Mons and the ejection sequence)

01 · THE MYSTERY

The mystery: how did the two moons of Mars — Phobos and Deimos — form on either side of the corotation line?

Mars has two small, light and remarkably porous moons: **Phobos** (mean diameter about 22 km, density 1,853–1,887 kg/m³) and **Deimos** (diameter about 12 km, density 1,471 kg/m³) — a third to a half of their volume is empty space. Both orbit Mars almost exactly above its equator (i = 1.09° and 1.8°) in almost perfectly circular orbits (e = 0.0151 and 0.0003), but on either side of **the corotation line** (today 20,428 km):

1. **Two moons with opposite fates (A.1):** Phobos (9,375 km, orbital period 7.65 hours — three orbits per Martian day) is *inside* the corotation line, so the tide pulls it down and it is falling toward Mars; Deimos (23,463 km, orbital period 30.3 hours) is *beyond* the corotation line, so the tide pushes it outward and it is receding.

2 **Utterly different faces (A.2):** Deimos is smooth and rounded, without grooves and with shallow, worn craters; Phobos is rough and grooved with systems of long parallel grooves, chains of pits, and the crater Stickney (9 km) at its blunt end.

3 **The limits of the mainstream theories (A.4):** asteroid capture cannot circularise two orbits into the equatorial plane; while a giant impact or ring–moon cycles involve foreign material (30%–65%) and strong melting that shifts volatile isotopes (zinc and potassium). In Paper 10 the LEF model presents a unified alternative: **young Mars grew a "tapered protrusion" on its equator, and an ordinary volcano near its tip — Pavonis Mons — released the two moons one after the other with no foreign body at all!**

02 · AN IMAGE

An image: a convex cone on the equator, a right triangle lying down, and a cork in the vent of a weightless volcano

Three simple images explain how a volcano on young Mars released two moons into orbit:

- **A pointed bulge against a planetary "pear" (C.1):** a young, hot planet rich in water and gases ($R = 1.5 \times R_{\text{today}} = 5,084$ km, $\rho = 1,166$ kg/m³) spun fast (7.3–8.3 hours per day). This rate is too moderate to deform the whole planet into a "pear" (which requires a day shorter than 5.8 hours), but strong enough to push magma and water toward the equator. Where the greatest mass accumulated, **a local pressure-driven bulge** rose, cooled, stiffened and was built from inside into a convex cone with a narrow tip — while the core of Mars hardly moved!
- **A right triangle that lay down westward (C.4–C.6):** while the bulge stood upright on the equator, its tip lagged behind against the direction of Mars' rotation (eastward) and so leaned **westward**. When Mars slowed, cooled and separated into layers, the bulge did not collapse suddenly but **lay down slowly westward** — leaving **the Tharsis plateau**, with a western slope 2–3

times steeper than the gentle eastern slope! The length measured today on the surface from the base at Valles Marineris to the tip near Pavonis (2,996 km, or 4,800–4,920 km on the young, flattened Mars) is **the hypotenuse L of a right triangle**, whose standing height $h = \sqrt{L^2 - b^2}$ carried Pavonis to a height of **4,450–4,610 km** above the equator!

- **A cork in a volcano's vent at zero-weight height (D.2–D.5):** at 9,880–10,170 km from the centre of Mars, gravity dropped to a quarter of its value at the surface (0.41–0.44 m/s²), and the centrifugal push of the rotation cancelled it completely (**zero net weight!**). There, an ordinary volcano with a shallow chamber 5 km deep and a routine Vulcanian overpressure (10–50 MPa, as in the Soufrière Hills eruptions on the Earth) pushed out two "corks" at 110–250 m/s — and the rotation of Mars (2.08–2.43 km/s) immediately gave them orbital speed!

03 · SCIENCE AND MODEL

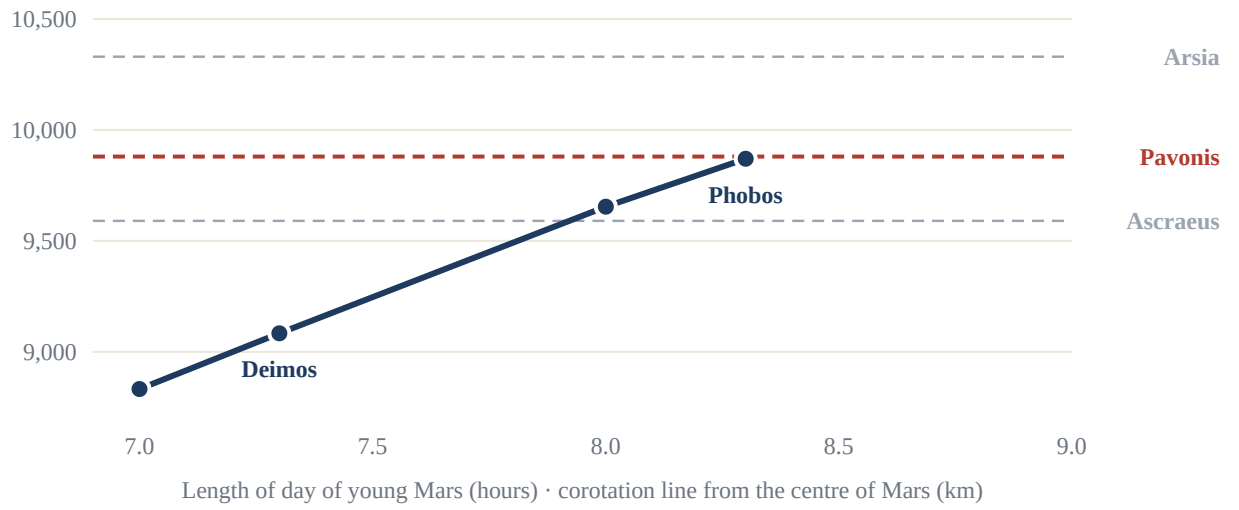
Science and model: the geometry of Tharsis, Pavonis Mons and the ejection sequence (Parts B–E)

Parts B–E of Paper 10 anchor each stage in the topography (MOLA), gravity (GMM-3) and orbital-mechanics data:

1. **The signature of the bulge in the topography and gravity field of Mars (C.3–C.9):** the bulge stretches along the equator (5°–12.5°S) from its base at Valles Marineris (294.7°E, 12.5°S, base half-width b = 1,340 km today and 1,280–1,450 km in its youth) to its tip near Pavonis (243.9°E, 5.4°S). In the 10°S–10°N band the western slope is steep (5.8 m/km over 1,360 km) against the gentle eastern one (2.0 m/km over 3,900 km). In the GMM-3 gravity field, **the long axis of the Martian equator points exactly to 255°E** (on the bulge!), the broad geoid peak (degrees 2–3) sits at the tip of the bulge (6.5°N, 253.5°E), and the overall shape of Mars is still offset by **1.37 km toward 262.7°E** — that is, **Mars still carries the remnant of the bulge!** Mars meets 9 of the 10 diagnostic criteria by direct observation, and the tenth follows from the model.

2. **Why Pavonis Mons and not Arsia, Ascraeus or Olympus? (D.3):** an orbital calculation for all four volcanoes shows that Olympus does not sit on the bulge at all; Ascraeus is too low at 8.3 h (inward of the corotation line, so ejecta from it falls back); Arsia would have thrown Deimos into an extreme orbit (e = 0.66) and Phobos into an elliptical orbit (e = 0.19); and only **Pavonis Mons** (1.48°N, 247.04°E, at a height of 4,450 km on the bulge) meets the corotation line at exactly 8.3 hours and releases Phobos directly into a circular orbit (e = 0.002) with no additional push!

3. **The two ejections and the 87% slowdown of Mars (E.1–E.3):**
    - 4. **First ejection — Deimos (T = 7.3 h, corotation line at 9,083 km):** the vent of Pavonis stood 1,090 km *beyond* the corotation line (sideways speed 2.43 km/s). The upper cork (light and rich in gas and water, ~ 20.7 km across) was thrown into a long elliptical orbit (10,170–23,990 km, e = 0.41, inclination ~ 10°) that crossed the ring on every orbit.
    - 5. **Second ejection — Phobos (T = 8.3 h, corotation line at 9,870 km):** after an hour of slowing in the rotation, the corotation line reached exactly to the vent of Pavonis (2.08 km/s). The second cork (larger, denser and drier, 22.9 km across) was released straight into a circular orbit (9,877–9,919 km, e = 0.002) at the inner edge of the ring!



The corotation line climbs as Mars slows: at 8.3 hours it meets exactly the vent of Pavonis (9,880 km from the centre of Mars).

04 · THE MECHANISM IN WORDS

The mechanism in words: reconstructing the triangle of the standing bulge and the Vulcanian push in the vent

Parts C.6, D.2–D.5 and E.2 of Paper 10 show how simple right-triangle geometry and the mechanics of an ordinary Earth-style Vulcanian explosion are enough to place two moons in orbit around Mars:

1. **Reconstructing the standing height from the lying hypotenuse (§C.6):** the length of 2,996 km measured today from the eastern end of the base at Valles Marineris (294.7°E, 12.5°S) to the tip near Pavonis (243.9°E, 5.4°S) is not the height of the bulge but **the hypotenuse L** of a right triangle drawn from the edge of the base to the tip above the centre of the base (272°E, 10°S, half-width b = 1,340 km today). On the young, flattened Mars ($R_{\text{mean}} = 5,084$ km), east–west lengths on the equator were enlarged by 1.60–1.64 times ($R_{\text{eq}} = 5,431–5,561$ km, so that L = 4,800–4,920 km), while the north–south width was compressed in the ratio c²/a (b = 1,280–1,450 km). From this the height of the standing bulge follows: $h = \sqrt{L^2 - b^2} = 4,580–4,740$ km (and in today's dimensions: 2,680 km)!

2 **The zero-net-weight condition and a routine Vulcanian push (§D.2, §D.5):** at the top of the bulge stood the vent of Pavonis, 4,450–4,610 km above the equator (9,880–10,170 km from the centre of Mars). There gravity dropped to a quarter of its value at the surface (0.41–0.44 m/s²), and the centrifugal push of the rotation cancelled it (net weight −0.17 m/s² at 7.3 h and 0.00 m/s² at 8.3 h). In these weightless conditions, gas that accumulated in a shallow chamber ~ 5 km deep at a routine Vulcanian overpressure of **10–50 MPa** pushed the cork along its own length at an exit speed of $v \approx \sqrt{2\Delta p/\rho} = 110\text{–}250$ m/s — while the rotational speed of the vent itself (2.08–2.43 km/s) supplied all the orbital speed!

3 **The budget for braking the rotation of Mars (§E.2):** a young, homogeneous Mars (I = 0.4 M R²) spinning in 7–9 hours carried an angular momentum **6.8–8.7** times its value today (I = 0.3644 M R², a day of 24.62 hours), and would even have sped up 2.5 times had it shrunk without losing momentum. Ejecting **6%–11% of the mass of Mars** (20%–33% of the water of a young Mars rich in over 30% water) through the corotation line (9,080–20,428 km) carried away 87% of the rotation and braked it almost to today's rate even before the ring dried out (8.3 → 22.3 h ejected 6%–9% of the mass, and 22.3 → 24.6 h only 0.1%).

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The standing height of the bulge h, the ejection speed from the vent under overpressure Δp, and the stability condition of a rotating homogeneous planet (a Maclaurin spheroid) are formulated as follows (§B.3–B.4, §C.6, §D.5):

$$h = \sqrt{L^2 - b^2} = 4{,}580\text{–}4{,}740 \text{ km}$$

$$v_{\text{exit}} \approx \sqrt{\frac{2\Delta p}{\rho}} = 110\text{–}250 \text{ m/s}$$

$$\frac{\Omega^2}{\pi G\rho} = 0.181\text{–}0.234 < 0.374 \quad (T_{\text{limit, Jacobi}} = 5.77\ \text{h},\ T_{\text{breakup}} = 5.27\ \text{h})$$

Since the dimensionless rotation parameter $\Omega^2/(\pi G\rho)$ for a day of 7.3–8.3 hours and a density of $\rho = 1,166$ kg/m³ equals 0.181–0.234 (below the triaxial-ellipsoid threshold of 0.374 and the break-up threshold of 0.449), young Mars kept global rotational stability and moderate flattening (0.18–0.24) and grew a local pressure-driven bulge instead of splitting into two large bodies.

06 · THE SIGNATURE

The signature in reality: the gravity map, the topography and the structural geology of Tharsis

Besides the match of the orbits of Phobos and Deimos, the bulge left four measurable geophysical signatures on the surface of Mars today (§C.5–C.8):

- **The free-air gravity gradient (GMM-3, degrees 2–90):** the free-air gravity rises continuously from **−565 mGal** at the eastern end of the base at Valles Marineris, through **+49 mGal** at the centre of the base (272°E, 10°S) and **+393 mGal** at the tip of the bulge (243.9°E, 5.4°S), up to **+1,474 mGal** at Pavonis, **+1,661 mGal** at Arsia and **+2,307 mGal** at Ascraeus.
- **The focus of radial faults (Anderson et al. 2001):** of 24,452 grabens and ridges around Tharsis across five geological periods (from the Noachian to the Amazonian), the centre of the radial faults migrated along the bulge from the early Claritas and Syria centres to **the Pavonis Mons area** in the last stage — and tectonic activity peaked at the beginning of Mars' history and waned with time.

The lab box: the parameters of young Mars, the standing bulge and the volcanoes of Tharsis (Paper 10, Parts B–E)

Length of day (T)	Flattening	Equatorial / polar radius (km)	Corotation line from the centre of Mars / above the equator (km)	The event in the Mars–moons system

7.0 hours	0.26	5,618 / 4,164	8,832 / 3,214	Building of the bulge and the early ring
7.3 hours	**0.24**	**5,561 / 4,250**	**9,083 / 3,522**	**Ejection of Deimos (a = 10,170–23,990 km, e = 0.41)**
8.0 hours	0.19	5,461 / 4,407	9,654 / 4,193	The corotation line climbs to the vent of Pavonis
8.3 hours	**0.18**	**5,431 / 4,455**	**9,870 / 4,439**	**Ejection of Phobos directly into a circle (e = 0.002)**
22.0 hours (~ 3.9 Ga)	≈ 0.01	≈ 3,410	≈ 19,000 / ≈ 15,600	The ring dries out and Phobos' "leash" is released
24.62 hours (today)	1/191	3,396 / 3,376	20,428 / 17,032	Phobos falls (9,375 km), Deimos recedes (23,463 km)

Volcano in Tharsis	Height on the bulge / distance from the centre of Mars (km)	Day at which the corotation line passes the vent	Ejection result at 7.3 h (Deimos)	Ejection result at 8.3 h (Phobos)
Ascraeus Mons	4,120 / 9,590	7.9 hours	Orbit (e = 0.25)	**Falls back** (the vent is inward of the line!)
Pavonis Mons	**4,450 / 9,880**	**8.3 hours**	**Elliptical orbit (e = 0.41)**	**An exact circle on the line (e = 0.002)**
Arsia Mons	4,950 / 10,330	8.9 hours	Extreme orbit (e = 0.66, up to 52,400 km)	Elliptical orbit (e = 0.19)
Olympus Mons	Off the bulge (5,600 from the centre)	—	**No orbit** (returns as dust)	**No orbit**

07 · NEXT

The door ahead

In this chapter we saw how young Mars, expanded and fast (R = 5,084 km, a day

of 7.3–8.3 hours), grew the pointed bulge of Tharsis on its equator, carried Pavonis Mons to a height of 4,450 km into zero-net-weight conditions, and with routine Vulcanian pressure released **Deimos** at 7.3 hours and **Phobos** at 8.3 hours.

But how did **the condensed steam ring** around Mars turn Deimos into a smooth, distant ball and Phobos into a grooved body that is falling toward Mars today — and how did the bulge drain its contents eastward through Valles Marineris? In the next chapter — **Chapter 20: The Steam Ring of Mars, the Drainage through Valles Marineris and the MMX Mission Test** (from Paper 10 — Parts F–O and Appendix B) — we will complete the whole story and the laboratory predictions for 2031!

Chapter 20

The Steam Ring of Mars, the Drainage through Valles Marineris and the MMX Mission Test →

← Chapter 18 · Prologue

© 2026 Zohar Duran · The LEF Model · lef-model.com · ORCID 0009-0005-3194-9387